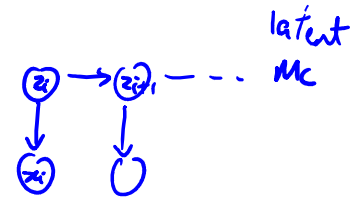
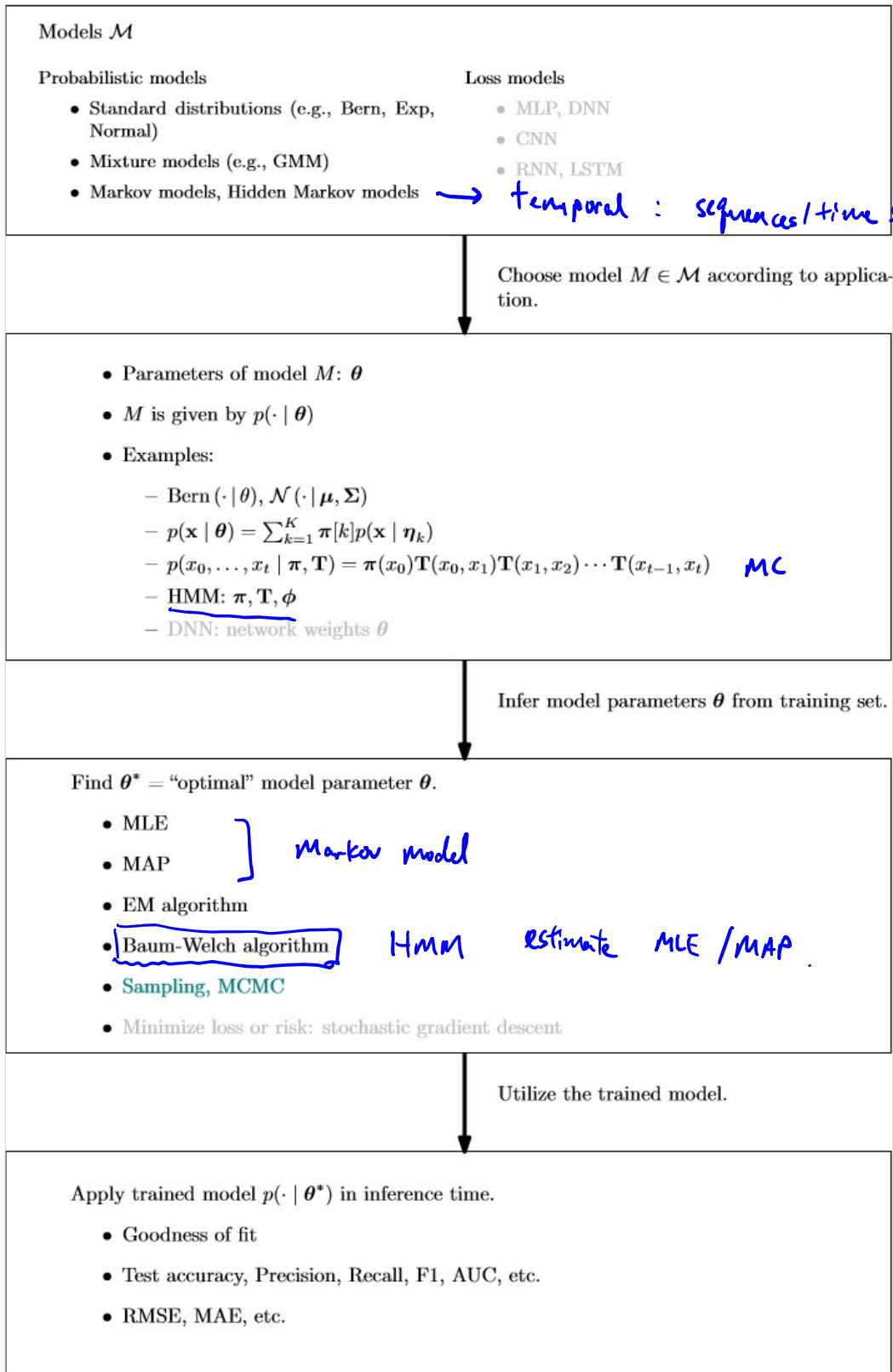




overview
week 5



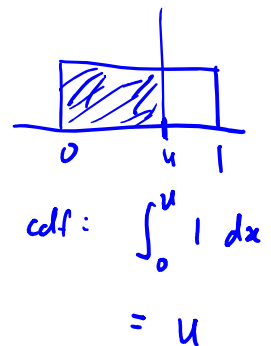
$$p(z|y) \stackrel{\text{Bayes' rule}}{=} \frac{p(y|z)p(z)}{p(y)}$$

Conjugate prior $p(z) \rightarrow p(z|y)$ comes from same distribution family as $p(z)$

cdf $F(x) = \mathbb{P}(X \leq x)$ Known $\quad$ Find $F^{-1}(\cdot)$

$$U \sim \text{Unif}[0, 1] \quad X = F^{-1}(U)$$

$$\begin{aligned} \mathbb{P}(X \leq x) &= \mathbb{P}(F^{-1}(U) \leq x) \\ &= \mathbb{P}(U \leq F(x)) \\ &= F(x) \end{aligned}$$



rand

$$X \sim \text{Exp}(\lambda) \quad \lambda e^{-\lambda x}, \quad x \geq 0$$

$$F(x) = \int_0^x \lambda e^{-\lambda x} \, dx = -e^{-\lambda x} \Big|_0^x = 1 - e^{-\lambda x}$$

$$u = 1 - e^{-\lambda x}$$

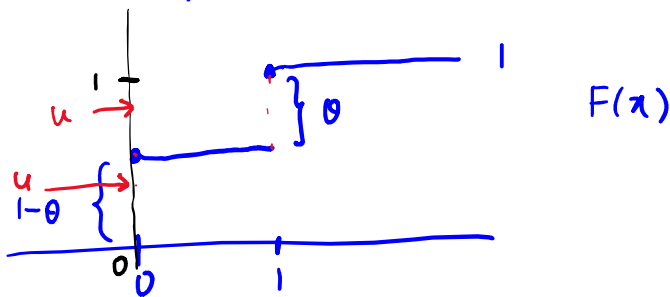
$$e^{-\lambda x} = 1 - u$$

$$x = -\frac{1}{\lambda} \log(1 - u) \quad F^{-1}(u) = -\frac{1}{\lambda} \log(1 - u)$$

$$U \sim \text{Unif}[0, 1], \quad F^{-1}(U) = -\frac{1}{\lambda} \log(1 - U)$$

$$U \sim \text{Unif}[0,1] \quad , \quad F^{-1}(u) = -\frac{1}{\lambda} \log(1-u)$$

$$X \sim \text{Bern}(\theta) \quad \mathbb{P}(X=1) = \theta \quad \mathbb{P}(X=0) = \underline{1-\theta}$$



$$0 \leq u \leq 1-\theta \quad F^{-1}(u) = 0$$

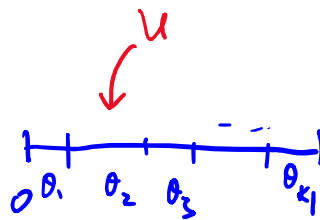
$$1-\theta < u \leq 1 \quad F^{-1}(u) = 1$$

$$U \sim \text{Unif}[0,1] \quad \rightarrow \text{prob. } 1-\theta$$

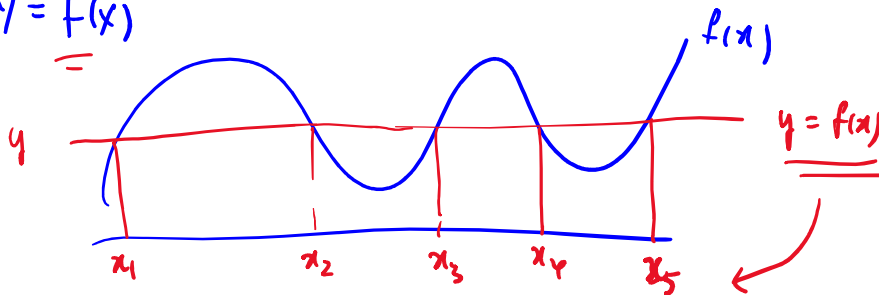
$$X = \begin{cases} 0 & \text{if } 0 \leq u \leq 1-\theta \\ 1 & \text{if } 1-\theta < u \leq 1 \end{cases}$$

$$\downarrow \text{prob. } \theta$$

$$\text{Cat}(\theta_1, \theta_2, \dots, \theta_K)$$



$$y = f(x)$$



$$\textcircled{P_Y(y)} = \sum_{i=1}^K \frac{P_X(x_i)}{|f'(x_i)|}$$

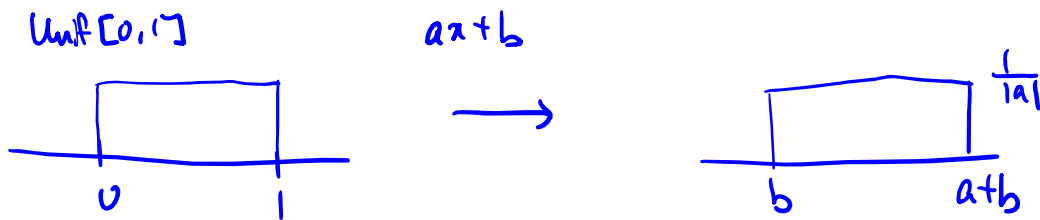
$$a \neq 0$$

$$y = ax + b \quad f(x) = ax + b \quad \underline{\underline{f'(x) = a}}$$

$$y = ax + b \Rightarrow \underline{\underline{x = \frac{y-b}{a}}}$$

$$y = ax + b \Rightarrow x = \underline{\underline{\frac{y-b}{a}}}$$

$$P_Y(y) = \frac{P_X\left(\frac{y-b}{a}\right)}{|a|}$$



$$y = x^2 \quad f(x) = x^2 \quad f'(x) = 2x$$

$$x^2 = y \rightarrow \begin{aligned} x_1 &= -\sqrt{y} \\ x_2 &= +\sqrt{y} \end{aligned}$$

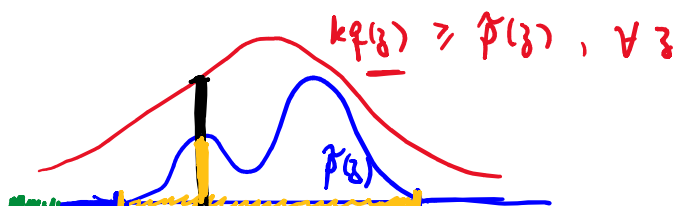
$$\begin{aligned} P_Y(y) &= \frac{P_X(-\sqrt{y})}{|2(-\sqrt{y})|} + \frac{P_X(+\sqrt{y})}{|2(+\sqrt{y})|} \\ &= \frac{1}{2\sqrt{y}} (P_X(-\sqrt{y}) + P_X(\sqrt{y})) \end{aligned}$$

Rejection Sampling

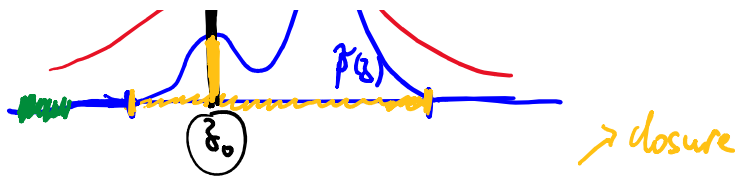
$$p(z) = \frac{1}{M} \tilde{p}(z) \quad M = \int_{-\infty}^{\infty} \tilde{p}(z) dz$$

target pdf

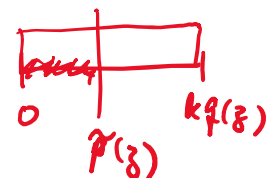
$q(z)$ - proposal pdf



$$\underline{u} \sim \underline{\text{Unif}[0, kq(\underline{z}_0)]}$$



$u \sim \text{Unif}[0, kq(z_0)]$
 if $u \leq \tilde{p}(z)$,
 accept.



$$\text{supp } p = \text{supp } \tilde{p} = \{z : p(z) > 0\}$$

$$\text{supp } p \subseteq \text{supp } q$$

$$z \sim q(\cdot)$$

$$P(z \text{ accepted}) = \int_{-\infty}^{\infty} \underbrace{P(z \text{ accepted} | z)} q(z) dz$$

$$= \int_{-\infty}^{\infty} \boxed{\frac{\tilde{p}(z)}{kq(z)}} q(z) dz$$

$$= \frac{1}{k} \int_{-\infty}^{\infty} \tilde{p}(z) dz$$

$$= \frac{M}{k}$$

$$kq(z) \geq \tilde{p}(z) \quad \forall z.$$

$$k \geq \frac{\tilde{p}(z)}{q(z)}, \quad \forall z$$

$$k = \max_z \frac{\tilde{p}(z)}{q(z)}$$

Prove that cdf of accepted z is target cdf = $\int_{-\infty}^{z_0} p(z) dz$

$$\downarrow$$

$$P(z \leq z_0 | z \text{ accepted})$$

$$\rightarrow \int \int \frac{1}{kq(z)} p(z) du dz$$

$$\mathbb{1}(\delta \leq \delta_0 \mid \delta \text{ accepted})$$

$$= \frac{\mathbb{P}(\delta \leq \delta_0, \delta \text{ accepted})}{\mathbb{P}(\delta \text{ accepted})}$$

$\frac{M}{K}$

$$= \left(\frac{1}{M}\right) \int_{\delta \leq \delta_0} \tilde{p}(\delta) d\delta$$

$$= \int_{\delta \leq \delta_0} p(\delta) d\delta \quad \text{target cdf}$$

$$\int_{\delta \leq \delta_0} \int_{u \in [0, \frac{1}{Kq(\delta)}]} \frac{1}{Kq(\delta)} p(\delta) du d\delta$$

$$\frac{1}{K} \int_{\delta \leq \delta_0} \left(\int_0^{\tilde{p}(\delta)} du \right) d\delta$$

$$= \frac{1}{K} \int_{\delta \leq \delta_0} \tilde{p}(\delta) d\delta$$

Sample target pdf $p(\delta)$

↑ normalization $\leadsto$ intractable
unnormalized $\tilde{p}(\delta)$

$q(\cdot) \rightarrow$ easy to sample, eg. unif, Gaussian, Exp, ---

$$K = \max_{\delta} \frac{\tilde{p}(\delta)}{q(\delta)}$$

$\delta \in \text{supp } p$

$$Kq(\delta) \geq \tilde{p}(\delta)$$

- Sample $\delta \sim q(\cdot)$
- sample $u \sim \text{unif}[0, Kq(\delta)]$
- Accept δ if $u \leq \tilde{p}(\delta)$

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) p(\theta)}{p(\mathcal{D})}$$

$$p(\theta | \mathcal{D}) = \frac{p(\mathcal{D} | \theta) p(\theta)}{p(\mathcal{D})}$$

unnormalized $\tilde{p}(\theta | \mathcal{D}) = p(\mathcal{D} | \theta) p(\theta)$

$$\frac{\tilde{p}(\theta)}{p(\theta)} = p(\mathcal{D} | \theta) \rightarrow \text{likelihood}$$

$$\hat{\theta} = \max_{\theta} \frac{\tilde{p}(\theta)}{p(\theta)} = \max_{\theta} p(\mathcal{D} | \theta) \rightarrow \text{MLE}$$

Importance Sampling

$$\mathbb{E}_{z \sim p(\cdot)} [f(z)] \stackrel{\text{SLLN}}{\simeq} \frac{1}{n} \sum_{i=1}^n f(z_i), \quad z_i \stackrel{\text{i.i.d.}}{\sim} p(\cdot)$$

"

$$\int_{-\infty}^{\infty} f(z) p(z) dz$$

$$f(z) = 1_E(z) = \begin{cases} 1 & \text{if } z \in E \\ 0 & \text{o.w.} \end{cases}$$

$$\begin{aligned} \mathbb{E}[f(z)] &= \mathbb{E}[1_E(z)] = \int_{-\infty}^{\infty} 1_E(z) p(z) dz \\ &= \int_E p(z) dz \\ &= \mathbb{P}(z \in E) \end{aligned}$$

target
 $p(z) \rightarrow$ difficult to sample

proposal
 $f(z) \rightarrow$ easy to sample.

$$\mathbb{E}_{z \sim p}[f(z)] = \int_{\mathbb{R}} f(z) p(z) dz$$

$$= \int_{\mathbb{R}} f(z) \underbrace{\frac{p(z)}{q(z)}}_{w(z)} q(z) dz$$

$$= \boxed{\int_{\mathbb{R}} f(z) w(z)} \boxed{q(z) dz} \quad \mathbb{E}_{z \sim q}[f(z) w(z)]$$

$$\simeq \frac{1}{n} \sum_{i=1}^n f(z_i) \underbrace{w(z_i)}_{\text{|| } \frac{p(z_i)}{q(z_i)}}, \quad z_i \stackrel{\text{i.i.d.}}{\sim} q(\cdot)$$

$$w(z_i) = \frac{p(z_i)}{q(z_i)}$$

↓

$$\text{|| } \frac{p(z_i)}{q(z_i)}$$

$$\frac{1}{n} \sum_{i=1}^n \underline{w(z_i)} f(z_i)$$

$$w_{\text{norm}}(z_i) = \frac{w(z_i)}{\sum_{j=1}^n w(z_j)}$$

$$= \frac{1}{n} \sum_{j=1}^n w(z_j) \sum_{i=1}^n w_{\text{norm}}(z_i) f(z_i)$$

$$n \rightarrow \infty$$

$$\rightarrow \mathbb{E}_q[w(z)] = \int_{\mathbb{R}} w(z) q(z) dz$$

$$= \int_{\mathbb{R}} \frac{p(z)}{q(z)} q(z) dz$$

$$= \int_{\mathbb{R}} p(z) dz$$

$$= 1$$

$$n \rightarrow \infty$$

$$\simeq \sum_{i=1}^n w_{\text{norm}}(z_i) f(z_i)$$

$$w_{\dots}(z_i) = \frac{w(z_i)}{\sum_{j=1}^n w(z_j)} = \frac{\tilde{p}(z_i)}{\tilde{q}(z_i)}$$

$$w_{\text{norm}}(z_i) = \frac{w(z_i)}{\sum_j w(z_j)} = \frac{\frac{p(z_i)}{q(z_i)}}{\sum_j \frac{p(z_j)}{q(z_j)}}$$

Sampling Importance Resampling (SIR)

- $z_1, z_2, \dots, z_n \sim q(\cdot)$
 - $w_{\text{norm}}(z_i)$, $i=1, 2, \dots, n$ $\rightarrow$ probabilities of z_i
 - Resample with replacement from $\{z_1, z_2, \dots, z_n\}$
 $\downarrow \quad \downarrow \quad \downarrow$
 $w_{\text{norm}}(z_1) \quad w_{\text{norm}}(z_2) \quad w_{\text{norm}}(z_n)$
- $\rightarrow$ output: $\tilde{z}$

$$\mathbb{P}(\tilde{z} \leq a \mid z_1, z_2, \dots, z_n)$$

$$= \sum_{i: z_i \leq a} w_{\text{norm}}(z_i)$$

$$= \frac{\frac{1}{n} \sum_i \mathbb{1}_{\{z_i \leq a\}} \frac{p(z_i)}{q(z_i)}}{\boxed{\frac{1}{n} \sum_j \frac{p(z_j)}{q(z_j)}}}$$

$$\xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}} \frac{p(z)}{q(z)} q(z) dz = 1$$

$$\xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}} \mathbb{1}_{\{z \leq a\}} \frac{p(z)}{q(z)} q(z) dz = \int_{\mathbb{R}} \mathbb{1}_{\{z \leq a\}} p(z) dz$$

$z_j \text{ i.i.d. } \sim q(\cdot)$

$$\int_{-\infty}^a p(z) dz$$

$$\parallel$$

 cdf of $p(\cdot)$

Bayesian:

$$\tilde{p}(\theta) = p(\mathcal{D}|\theta) p(\theta)$$

$$q(\theta) = p(\theta)$$

$$w_{\text{norm}}(\theta_i) =$$

$$\mathcal{J} = \left[\begin{array}{c} \vdots \end{array} \right] \left\{ \begin{array}{c} 256^2 \end{array} \right.$$

$$\begin{array}{c} 256 \\ 256 \end{array} \left[\begin{array}{c} \end{array} \right]$$